PCA on High-Dimensional Human Activity Recognition (HAR)

This report accompanies the *pca-har-KNN* project: a KNN baseline on the UCI HAR dataset (561 features) compared with KNN after PCA-based dimensionality reduction.

1. What is PCA? (Simple Explanation)

Principal Component Analysis (PCA) finds a new set of axes for your data. The first axis (first *principal component*) is the direction along which the data varies the most. The second axis captures the next highest variation and is orthogonal (independent) from the first, and so on.

You can use PCA to:

- **Visualize** high-dimensional data in 2D or 3D by keeping only the first few components
- **Compress** the data by keeping only the most important components that explain most of the variance

This is especially useful for models like KNN because it can:

- Speed up computation
- Reduce noise

Important: PCA is *unsupervised*, meaning it does not use labels—only feature values (after scaling).

2. Why PCA Helps with High-Dimensional Data

Real-world datasets (like HAR sensor data) often contain many **correlated features**, which leads to:

Problems:

- **Redundancy:** Multiple features carry similar information
- **Curse of dimensionality:** Distance calculations become less meaningful in high dimensions

How PCA Fixes This:

PCA transforms the original features into a smaller set of **uncorrelated components**, ordered by importance (variance).

Benefits:

- Reduces dimensionality (e.g., 561 → ~100 for 95% variance)
- Speeds up KNN predictions
- Removes noise by ignoring low-variance components

Trade-Off:

- You might lose small but important details
- Accuracy may:
  - Stay the same
  - Improve slightly
  - Drop slightly

3. Eigenvalues and Eigenvectors (From Scratch)

For a square matrix $\mathbf{A}$, a non-zero vector $\mathbf{v}$ is an **eigenvector** if multiplying $\mathbf{A}$ by $\mathbf{v}$ only *scales* $\mathbf{v}$:

$$A\mathbf{v} = \lambda\mathbf{v}$$

The scalar λ is the **eigenvalue** associated with $\mathbf{v}$.

Example Matrix

Consider a matrix representing the covariance between features:

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

Characteristic Equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$

To find the eigenvalues, we solve for when the determinant of the matrix minus the identity scaled by λ is zero:

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 4-\lambda & 2 \\ 2 & 3-\lambda \end{bmatrix}$$

The determinant calculation:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - (2 \times 2) = 0$$

$$\lambda^2 - 7\lambda + 12 - 4 = 0$$

$$\lambda^2 - 7\lambda + 8 = 0$$

Solve for Eigenvalues

Using the quadratic formula $\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$\lambda = \frac{7 \pm \sqrt{(-7)^2 - 4(1)(8)}}{2(1)}$$

$$\lambda = \frac{7 \pm \sqrt{49 - 32}}{2}$$

$$\lambda = \frac{7 \pm \sqrt{17}}{2}$$

This gives us our two principal values:

- $\lambda_1 = \frac{7+\sqrt{17}}{2} \approx 5.56$
- $\lambda_2 = \frac{7-\sqrt{17}}{2} \approx 1.44$

Eigenvectors (Conceptually)

For each eigenvalue (λ), we solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0$ to find the direction of the vector $\mathbf{v}$.

Key Takeaways:

- **Eigenvalues** indicate the magnitude of variance (how much "stretching" occurs) along a specific axis.
- **Eigenvectors** define the principal directions (the new axes) of the data.

Connection to PCA:

- PCA uses the **covariance matrix** of the data
- Eigenvectors = principal components (directions)
- Eigenvalues = amount of variance in each direction

👉 In simple terms:

- Eigenvectors = directions of data spread
- Eigenvalues = how important each direction is

PCA selects the directions with the **largest eigenvalues.**

4. Results Interpretation

After running:

```
python main.py
```

Output Files:

- **results/metrics.json**
  - Accuracy, precision, recall, F1 score
  - Confusion matrices
  - Training and prediction time
- **results/plots/**
  - 2D PCA visualization
  - Explained variance graph
  - Accuracy comparison chart
  - Confusion matrix heatmaps

Before vs After PCA

Before PCA:

- Uses all **561 features**

After PCA:

- Keeps ~95% variance
- Typically reduces to around **100 components**

Trade-Off Comparison

Aspect	Without PCA	With PCA (~95% variance)
Dimensionality	High (561)	Much lower
Speed	Often slower prediction	Often faster prediction
Accuracy	Baseline	Often similar; may change slightly

Key Insight

PCA does NOT guarantee better accuracy.

Instead, it provides:

- **Massive dimensionality reduction**
- **Faster computation**
- **Better visualization (2D/3D plots)**